

BOKU Power BI Migration

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Problem Framing

LEGACY SYSTEM

- ▶ Oracle Warehouse Builder (ETL)
- ▶ SAP BusinessObjects (Reporting)
- ▶ 20+ years of accumulated complexity

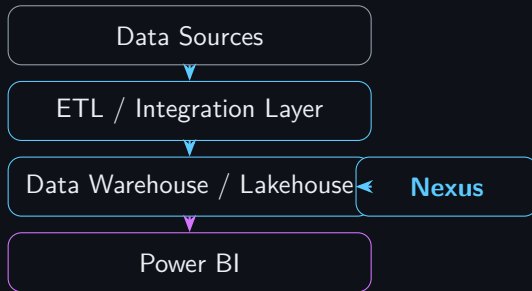
CHALLENGES

- ▶ Fragmented ETL and reporting layers
- ▶ Limited scalability and flexibility
- ▶ No path to analytics or AI integration

**Replace with a modern, scalable,
unified data platform**

Target Architecture

LAYERS



KEY PRINCIPLE

- ▶ Power BI = visualization layer only
- ▶ Logic and transformations live upstream
- ▶ Clear separation: Ingestion · Storage · Viz

Migration Strategy

PHASE 1 — INVENTORY

- ▶ Catalogue all reports and ETL jobs
- ▶ Classify: critical / replaceable / obsolete
- ▶ Map data sources and owners

PHASE 2 — PARALLEL BUILD

- ▶ New system runs alongside legacy
- ▶ Pilot reports validated against old outputs
- ▶ No big bang

PHASE 3 — CUTOVER

- ▶ Go-live per report cluster
- ▶ Rollback plan on standby
- ▶ Legacy decommission — 3-month buffer

Risks & Pitfalls

TECHNICAL

- ▶ Hidden dependencies in ETL processes
- ▶ Data quality and consistency gaps
- ▶ Semantic equivalence issues (same KPI, different formula)
- ▶ Performance differences post-migration

ORGANIZATIONAL

- ▶ 20 years of user habit to overcome
- ▶ Misaligned stakeholder expectations
- ▶ Training gap across teams

MITIGATION

- ▶ Automated validation: old vs. new value-by-value
- ▶ Key users in pilot group from day one
- ▶ Versioned datasets, no silent overwrites
- ▶ Usage triage: drop reports with fewer than 5 views/month

Role of Power BI

STRENGTHS

- ▶ Fast interactive dashboard creation
- ▶ Deep Microsoft ecosystem integration
- ▶ Self-service BI for business users

TECHNICAL MODES

- ▶ Import Mode — in-memory performance
- ▶ DirectQuery — live source connection
- ▶ Semantic modelling layer (DAX)

**Power BI is a visualization layer —
not the data warehouse**

Nexus Layer

CLASSICAL BI

Tables → Dashboards One domain · one tool · one query

NEXUS CONCEPT

Data → Typed Scientific Objects

- ▶ Time domain (kinetics, SPR)
- ▶ Reciprocal space (SAXS, $I(q)$)
- ▶ Real space ($p(r)$, geometry)

**Transition: Reporting →
Computational Science Platform**

Final Vision

FROM

- ▶ Static reporting (SAP BO dashboards)
- ▶ Manual ETL maintenance
- ▶ Isolated domains, no AI/ML readiness

TO

- ▶ Dynamic, scalable data platform
- ▶ AI-ready infrastructure
- ▶ Integrated scientific + administrative analytics
- ▶ Foundation for Nexus modelling

**Data Warehouse = Foundation for
Nexus**